

Machine learning notes

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1 Machine Learning Fundamentals

1.1 What is Machine Learning

Machine learning (ML) is a subset of Artificial Intelligence (AI) where the system automatically learns pattern from data and improves their performance without being explicitly programmed.

1.2 Supervised learning vs Unsupervised learning

Supervised learning trains models using labeled data to predict outcome.

Unsupervised learning finds hidden patterns and structures in unlabeled data.

1.3 Classification and regression

Classification predicts discrete categories or labels (e.g. spam vs. not spam).

Regression predicts continuous numerical values (e.g. predicting house prices)

1.4 Train / validation / test split

training set: used to train the model.

validation set: during training to tune hyper-parameters and prevent overfitting.

testing set: used only once for evaluation of the final model.

1.5 Overfitting vs Underfitting

overfitting: The model is too complex. It learns the training data very well, but fails on the testing/evaluation data.

underfitting: The model is too simple to understand the data, so it performs poorly on both the data it was trained on and the new data it is tested on.

1.6 Basic data preprocessing

a) missing value - drop, imputation

b) categorical data - label encoding, one hot encoding

c) feature scaling - normalization, standardization

1.normalization(min-max)

2.standardization(z-score)

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (1)$$

$$X_{\text{std}} = \frac{X - \mu}{\sigma} \quad (2)$$

d) outliers - z-score, IQR

2 Model Evaluation

2.1 Classification Metrics

Table 1: evaluation matrix

act pred	True	False
True	TP	FN
False	FP	TN

2.1.1 Definition

True Positive (TP): Correctly predicted positive.

True Negative (TN): Correctly predicted negative.

False Positive (FP): Incorrectly predicted positive (Type I Error).

False Negative (FN): Incorrectly predicted negative (Type II Error).

2.1.2 Accuracy

The ratio of correctly predicted observations to the total observations.

$$Accuracy = \frac{TP + TN}{TP + TN + FN + FP}$$

2.1.3 Precision

Out of all the instances the model predicted as positive, how many are actually positive?

$$Precision = \frac{TP}{TP + FP}$$

2.1.4 Recall

Out of all the actual positive instances, how many did the model correctly identify?

$$Recall = \frac{TP}{TP + FN}$$

2.1.5 F1-score

The harmonic mean of Precision and Recall. It provides a single metric that balances both.

$$F1 - score = 2 \cdot \frac{precision \cdot recall}{precision + recall}$$

2.1.6 ROC and AUC

The model first assigns a probability score to each sample. The samples are then sorted from highest to lowest score, and the algorithm iterates through them one by one. At each step, the True Positive Rate (TPR) and False Positive Rate (FPR) are calculated as coordinate points, which are connected to form a step-like ROC curve. This curve essentially serves as a "business decision menu," illustrating the model's performance across various thresholds and helping stakeholders scientifically define the final decision range based on their tolerance for false positives and false negatives. The total area under this ROC curve is the AUC (Area Under the Curve), which acts as the ultimate scorecard for the model's overall capability. Fundamentally representing the probability that the model ranks a randomly chosen positive sample higher than a randomly chosen negative one, AUC provides an objective measure of the model's discriminative power, remaining highly robust even in cases of extreme class imbalance

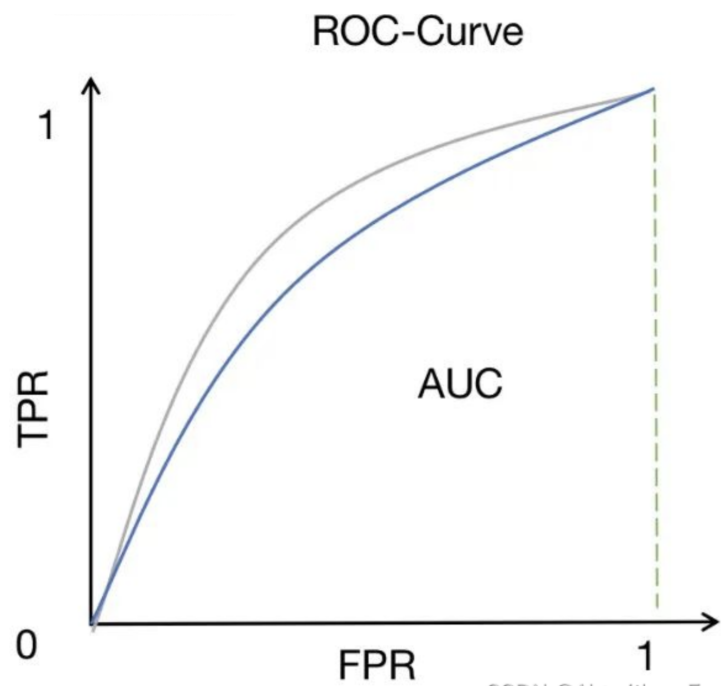


Figure 1: ROC And AUC